

ISABELLA CHANDLER

Biomedical Engineering | Medical Devices | Quality & Product Development
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Portfolio

PROFILE

Biomedical Engineering student with hands-on experience in device quality inspection, technical troubleshooting, biomedical prototyping, laboratory support, and project leadership. Quality Science Education Certificate training in FDA quality principles, GMP, validation, risk management, and documentation. Interested in medical devices, quality engineering, healthcare technology, testing, and product development.

EDUCATION & QUALITY TRAINING

B.S. Biomedical Engineering | Lawrence Technological University, Southfield, MI

Expected May 2027

GPA: 3.16/4.00

Current coursework: Tissue Engineering + Lab, Biomechanics + Lab, Foundations of Medical Imaging, Introduction to Nanotechnology, BME Projects I

Relevant foundation: Biotransport, Circuits & Electronics + Lab, Anatomy & Physiology + Lab, General/Organic/Biochemistry + Lab, Probability & Statistics

Quality Science Education Certificate | Pathway for Patient Health

Aug 2025

Training: FDA quality principles, GMP, validation, risk assessment, change control, data integrity, and quality systems for pharmaceutical and medical-device environments

EXPERIENCE

Help Desk Assistant | Lawrence Technological University

Jan 2025-Present

- Processed, inspected, cleaned, imaged, and prepared 2,000+ university laptops for redistribution, including Lenovo T1/T2/T3, x360, and Fujitsu systems; identified defects and verified device readiness against consistent quality expectations.
- Supported high-volume device deployment and independently maintained Summer 2026 back-room operations by organizing inventory, charging systems, preparing accessories, monitoring printer readiness, assisting technicians, and providing professional front-desk and phone support.

Biomedical Engineering Summer Camp Lab Assistant | Lawrence Technological University

Apr-Aug 2024

- Prepared BME teaching laboratories for high-school activities, organized equipment and storage areas, updated Excel inventory records, and completed routine lab-readiness tasks including eyewash-station cleaning.

Director of Programming | LTU Student Government

Sep 2024-May 2026

- Led planning and execution of the 2026 end-of-year celebration with a \$10,000+ budget, coordinating vendor quotes, food trucks, inflatables, decorations, indoor/outdoor activities, schedules, and campus stakeholders.
- Advanced from Student Senator to committee leader and Executive Board Director of Programming; maintained the event calendar, led meetings and presentations, evaluated funding/organization requests, and communicated professionally across campus groups.

SELECTED BIOMEDICAL ENGINEERING PROJECTS

Mopec Tissue Insight - Company-Sponsored Senior Project | BME Projects I

2026-Present

- Contributing to early-stage development of an AI-assisted gross-tissue imaging workflow through dataset review, image-quality considerations, preprocessing/annotation planning, train-validation-test strategy, technical documentation, and expert-review integration.

HeadUp Posture Feedback Wearable | Bioinstrumentation

- Designed and tested an accelerometer-based posture wearable with user calibration and threshold logic; triggered haptic/visual feedback after sustained forward deviation and evaluated sensing accuracy, delay, feedback strength, and comfort.

ConnectUs Accessible Dice-Tower Redesign | Entrepreneurial Engineering Design Studio

- Led an accessibility-focused redesign from user pain-point identification through decision matrices, three prototype iterations, testing, and final demonstration; integrated a push-button servo, circuitry, 3D-printed components, soldering/wiring, and mechanical troubleshooting.

TECHNICAL SKILLS

Quality & engineering: quality inspection, validation concepts, risk assessment, technical documentation, requirements tracking, prototyping, 3D printing, sensor calibration/testing, soldering/wiring, laboratory setup

Software & tools: MATLAB, C++, Python/CircuitPython, Arduino, BLE, COMSOL, AutoCAD/CAD, CircuitLab, Capstone, Excel, Gemini AI Studio, Mu Editor